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Editor-in-Chief
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Dear Editor,

We are pleased to submit our manuscript titled "The Conditioning Set Decides the Measured Transfer Asymmetry: An Exact Correction and a Matched-Pair Prediction for CNN versus Vision Transformer Robustness Comparisons" for consideration as a Regular Article in *IEEE Access*.

Studies that compare the transferability of adversarial examples between CNNs and Vision Transformers state which samples they score, the conditioning set, but rarely vary it, so its weight in the conclusion has not been measured. The manuscript measures it on fixed models: adversarially trained ResNet-18 and ViT-Tiny on CIFAR-10, three independent training runs per architecture, per-sample logging on the full test set and a validation split held out before any training. The absolute ordering is not in question, since the CNN leads under AutoAttack ($37.93 \pm 0.14\%$ against $29.14 \pm 0.40\%$, McNemar significant in every run); the paper examines the transfer asymmetry between the two families. Its contributions are the following.

1. With models, data and attack held fixed, four conditioning protocols place the CNN-to-ViT transfer asymmetry between +4.4 and +19.4 points (mean per-pair spread $15.01 \pm 0.84$ points), whereas retraining both architectures moves it across a range of at most 2.41 points within any protocol, with standard deviations of 0.23 to 1.27 points. Three separate facts describe this range, and they do not add up to it: the 9.21 points between the unconditioned and target-correct estimates, the successful-source protocol lying 11.10 points above both-correct because it estimates a different quantity, and 3.91 points (run-level 95% interval $[3.12, 4.70]$) between the two protocols we endorse.
2. The 9.21-point gap follows from the law of total probability, the transfer analogue of the TRADES decomposition. Its premise, the probability $p$ that a sample the target already misclassifies stays misclassified under the transferred perturbation, measures 0.989 to 1.000 on adversarially trained targets, where the closed-form correction is exact; on standard-trained targets $p$ falls to 0.877 and the $p = 1$ form misses by up to 1.58 points, so the general form with a measured $p$ is required.
3. The direction of the asymmetry is a source effect. Additive source-plus-target fits to the target-correct transfer matrices give source terms of +5.13 (CIFAR-10) and +9.54 points (CIFAR-100) against target terms of −2.03 and −6.19: examples crafted on our ViT-Tiny transfer worse to every target, a fixed WideResNet-28-10 included, while the ViT is the less vulnerable target. Applied to the published $8 \times 8$ matrices of Mahmood et al. (ICCV 2021), the same analysis shows that the conditioning protocol does not explain their opposite direction on CIFAR-10, and that their CIFAR-100 pairs sharing our direction reach it through the

target term; the manuscript states this disagreement as an open question.

4. The magnitude follows the clean-accuracy gap of the pair. The four-protocol spread falls to 3.70 points on SVHN, where the two models are 1.85 points apart in clean accuracy, and to 0.19 points for a standard-trained ResNet-18 and WideResNet-28-10 that are 0.47 points apart; across the 28 pairs of an eight-model pool built from RobustBench checkpoints, without any training of our own, the three-protocol spread rises by 0.872 points per point of gap. Re-scoring the same pairs at their last checkpoint moves the spread from 15.01 to 12.84 points without changing the direction, so the checkpoint-selection rule is a measurable part of the result.
5. Three readings that architecture comparisons often suggest do not hold under direct measurement: CNN gradients are sparser but not more spatially localized, CLS attention entropy does not change under attack (run-level equivalence in all 12 layers), and a ViT-specific transfer attack gives no gain over plain MI-FGSM at a matched budget. These negative results, and a pre-registered prediction that failed, are reported in full.

The manuscript closes with a numbered set of reporting requirements for cross-architecture robustness studies, the first two being to state the conditioning set of every transfer number and not to present unconditioned transfer rates as architecture comparisons. Every protocol estimate is recomputed from per-sample evaluation records, and the numbers in the text are checked against the released artifacts by scripts that are part of the release.

We believe this work suits *IEEE Access* because it is methodological rather than architecture-specific: its subject is how a widely used quantity is measured, and its correction applies to any study that compares transfer across model families. The journal's interdisciplinary readership and open-access model fit a contribution whose value lies in being adopted as reporting practice.

As stated in the Acknowledgment of the manuscript, the authors used Claude (Anthropic) to assist with drafting and revising the text and with implementing the analysis and verification scripts. The study design, the experimental protocol and all conclusions are the authors', who verified every reported number against the released artifacts. All figures are produced by plotting scripts or, for the study overview, as a vector schematic; none is produced by an image-generation model.

**Declarations:**

- This manuscript is original work and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- A preliminary subset of these results, limited to CIFAR-10, is under review as a conference paper (ATEEC 2026); the present manuscript is the extended version and cites it. Beyond the conference version, the journal version adds the CIFAR-100 and SVHN datasets, the $L_2$ threat model, the total-probability identity together with the measurement of its premise $p$, the additive source–target decomposition and its external test on the published matrices of Mahmood et al., standard-trained targets and an eight-model RobustBench pool, the analysis of the checkpoint-selection rule, the transformer-specific transfer attack and the pre-registered analysis protocol, and it corrects one claim of the conference version.
- All authors have reviewed and approved the final manuscript.
- All experiments use the publicly available CIFAR-10, CIFAR-100, and SVHN datasets and standard evaluation protocols.

- The authors declare no conflict of interest.
- The analysis pipeline, the scripts that generate every table and figure, the per-sample evaluation records (including the clean-trained pairs, the RobustBench pool and the last-checkpoint re-scoring), a machine-read copy of the published tables of Mahmood et al. with the scripts that re-score them, the fixed validation-split indices, and the seed lists are publicly available at `https://github.com/cagrisahin58/cnn-vit-robustness-protocols`. Trained checkpoints total about 40 GB and are available from the authors on request.

Thank you for considering our manuscript. We look forward to your editorial decision.

Sincerely,

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